

Lecture16: Dimension Reduction2

Robust PCA

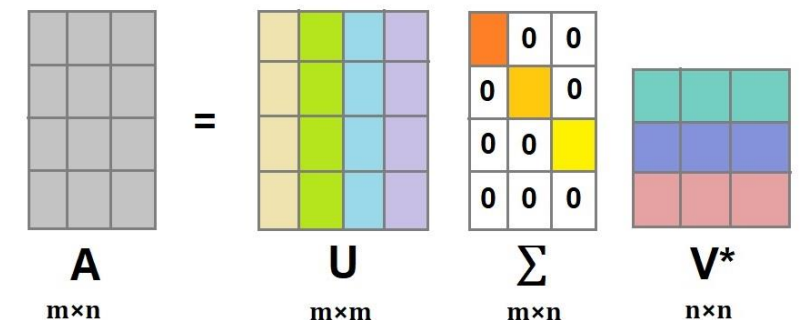
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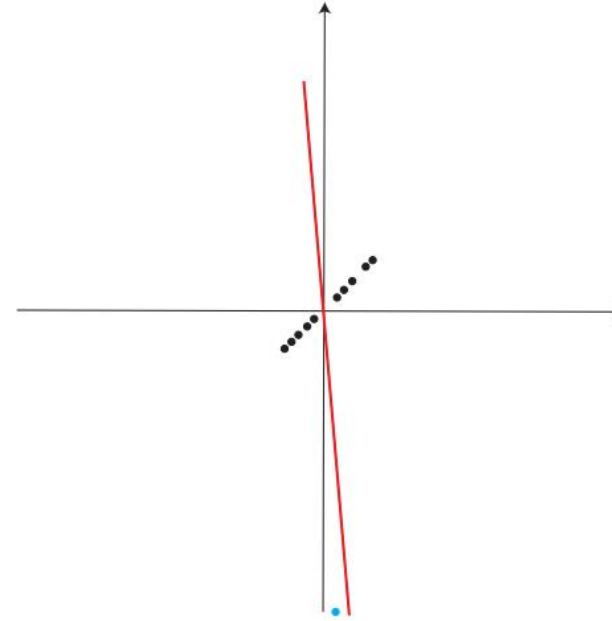
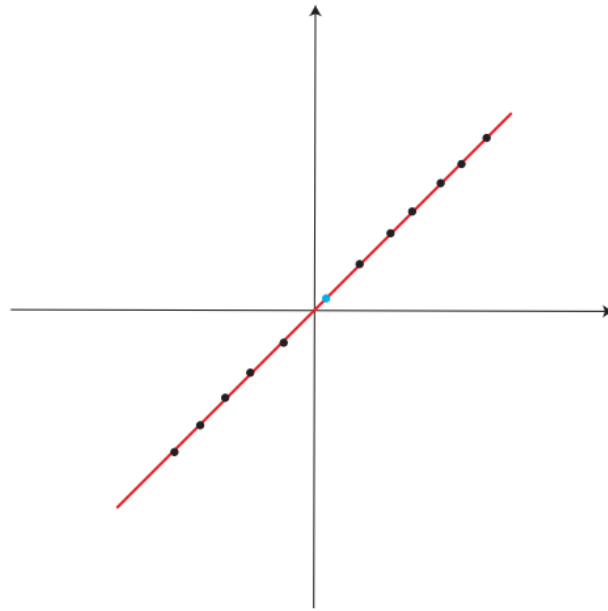
Low-Rank Approximation

- In the last class, we discussed Principal Component Analysis (PCA) and Singular Value Decomposition (SVD).
- Some math terminologies:
 - Rank: the number of non-zero singular values of a matrix.
 - Low-rank matrix: a matrix has only a few non-zero singular values.
- Truncated SVD (i.e., keeps only a few largest singular values) is a low-rank matrix approximation.
 - Finding the closest low-rank matrix to the original matrix.
 - $\min_{\tilde{X}} ||X - \tilde{X}||_F$ subject to $\tilde{X}$ is low – rank
 - Discuss Frobenius norm on board
- PCA $\approx$ Low-rank matrix approximation



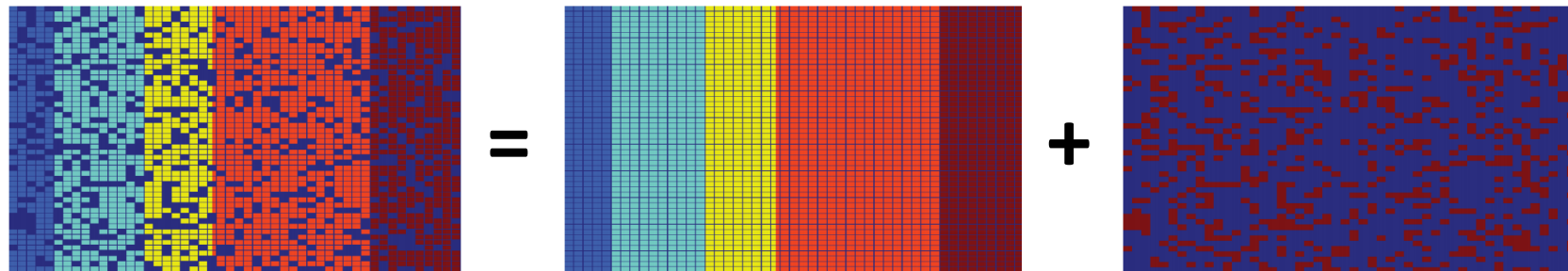
Principal Component Analysis

- Tolerates small noise very well, but is over-sensitive to outliers.
 - This is true for many other methods we have introduced this semester.
 - A common issue for ℓ_2 based loss function.
- One single extreme outlier can completely change the PCA result



Robust PCA

- Robust PCA (RPCA) is design to overcome this outlier issue
- Observed a corrupted data matrix $Y = X + S$
 - X is a low-rank matrix
 - S is a sparse matrix



Robust PCA

- Robust PCA (RPCA) is design to overcome this outlier issue
- Observed a corrupted data matrix $Y = X + S$
 - X is a low-rank matrix
 - S is a sparse matrix
- Recover the underlying X and S simultaneously from the observed Y :

$$\min_{\tilde{X}, \tilde{S}} ||\tilde{X} + \tilde{S} - Y||_F \quad \text{subject to } \tilde{X} \text{ is low - rank and } \tilde{S} \text{ is sparse}$$

- Tolerates outliers by putting them into the sparse matrix

Many Applications

- Video background subtraction
- Face modeling
- Image alignment
- Feature identification
- Community detection
- and many others . . .

Application Example: Video Background Subtraction

- Given a video with static background (e.g. from a security camera)
 - Each frame is vectorized and becomes a column of the data matrix
 - Underlying subspace (low-rank component): static background
 - Sparse outliers: foreground, i.e., moving objects



Watch Video Demos



Watch Video Demos



Who's Outliers?

- If someone stays in the same place all the time, he/she is part of the background.
- People who keep walking are obviously a foreground (outlier).
- How about the girl who stays there for a minute and then starts walking?

Who's Outliers?

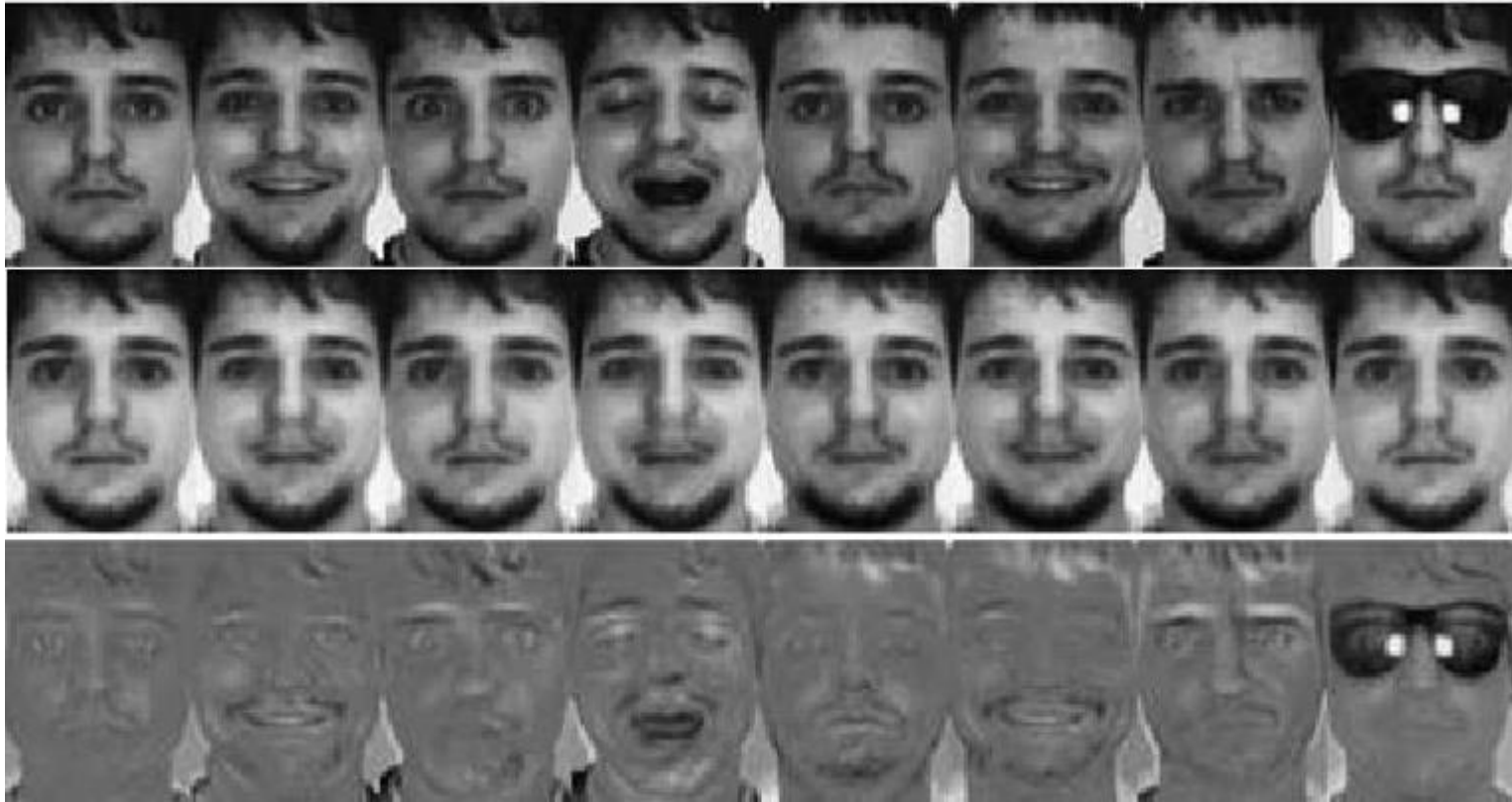
- If someone stays in the same place all the time, he/she is part of the background.
- People who keep walking are obviously a foreground (outlier).
- How about the girl who stays there for a minute and then starts walking?
- Rank (and some other parameters) determines the outliers in the “gray area”
 - The higher the rank is, the more change is allowed in the background. So the girl can be part of the background for the first minute, then become an outlier once she walks.
 - The lower the rank is, the more static the background is. The girl is treated as an outlier as long as she even moved.

Application Example: Face Modeling

- Collect many pictures of the same person
 - Underlying subspace (low-rank component): The true face
 - Sparse outliers: Shadow, mask, sunglasses
- A preprocessing step for face recognition



Face Modeling Demo



Time to Play

- Unfortunately, python doesn't have an official RPCA module.
- Let's learn how to use other people's code through GitHub!
- Make sure you watch every step very carefully—I am not directly showing the “right” steps but the right way to find the right steps.
- Use <https://github.com/sverdoot/robust-pca> as an example.



One More Thing

- One of my Robust PCA algorithms has only Matlab implementation on GitHub
 - Accelerated Alternating Projections for Robust Principal Component Analysis
 - https://github.com/caesarcai/AccAltProj_for_RPCA
- I am looking for a student to rewrite the code in Python
- The reward will be a good recommendation letter if you apply for graduate school